Biochemistry of Nutrient Digestion

- **ILOs**

1. **List the various enzymes participating in digestion processes of carbohydrates, proteins, lipids, and nucleic acid and mention their specific role and outcome of deficiency .**
2. **List the precursors of bile acids describe their synthesis and composition.**
3. **Explain the role of bile on digestion and functions for each of bile components.**
4. **Discuss the biochemical basis of gall bladder stones.**
5. **Illustrate biochemical basis and outcome of: Malabsorption syndromes, Lactose intolerance, Steatorrhea& Pancreatitis**

Absorbable Food Substances

Absorbable Food Substances	
Source	Substance
Carbohydrates	Monosaccharides: glucose, galactose, and fructose
Proteins	Single amino acids, dipeptides, and tripeptides
Triglycerides	Monoacylglycerides, glycerol, and free fatty acids
Nucleic acids	Pentose sugars, phosphates, and nitrogenous bases

1-Carbohydrate Digestion

- 1-starch and glycgen ---amylase enzyme---► maltose
- 2-sucrose ---sucrase---► glucose + fructose
- 3-lactose ---lactase---► glucose + galactose
- 4- maltose ---maltase---► glucose + glucose

Enzymes for cabohydrates digestion

- **1-Amylase enzyme: two types**

a-Salivary amylase: secreted from Salivary glandsbreaks down polysaccharides (glycogen and starch) into maltose and dextrins.

B-Pancreatic Amylase: The pancreas releases α-amylase into the small intestine,breaks down glycogen and starch into maltose and dextrins.

- **2-α-Dextrinase:** α-dextrinase (also known as isomaltase) is a crucial digestive enzyme located in the intestinal brush border membrane. It breaks down α-1,6 glycosidic bonds in dextrin.
- 3- Lactase, Maltase and Sucrase:intestinal brush border membrane enzymes breaks down disaccharides.

2-Protein Digestion

- **Proteins are polymers composed of amino acids linked by peptide bonds to form long chains. Digestion converts proteien to its constituent amino acids.**

Enzymes for protein digestion

- **1-pepsin:** **secreted by the chief cells of the stomach as pepsinogen and activated to pepsin by hydrochloric acid (HCl) to initiate protein digestion**
- **pepsin**
- **Protein ----------► peptides**

- **Trypsin and chymotrypsin:** both secreted by Pancreatic acinar cells as zymogens (trypsinogen and chymotrypsinogen) and activated in the small intestine from inactive zymogens to prevent premature pancreatic damage. Enteropeptidase activates trypsinogen to trypsin, which then acts as the master enzyme, activating chymotrypsinogen and other pancreatic enzymes, with calcium helping to stabilize the process.

3-Lipid Digestion

- **Lipid processing is emulsification, which suspends fat molecules to expose lipid molecule surface areas to hydrolytic enzymes. Emulsification starts in the mouth via mastication and lingual lipase and continues with gastric mixing.**

- **Further fat solubilization depends on raising the pH to approximately 6.5 and mixing fat digestion products with bile salts released by the gallbladder.The pancreas release lipolytic enzymes, and most lipid absorption occurs in the proximal two-thirds of the jejunum (ie, proximal small intestine).**

4-Nucleic Acid Digestion

- **The nucleic acids DNA and RNA are found in most of our food. Two types of pancreatic nuclease are responsible for their digestion: deoxyribonuclease, which digests DNA, and ribonuclease, which digests RNA. The nucleotides produced by this digestion are further broken down by two intestinal brush border enzymes (nucleosidase and phosphatase) into pentoses, phosphates, and nitrogenous bases, which can be absorbed through the alimentary canal wall.**

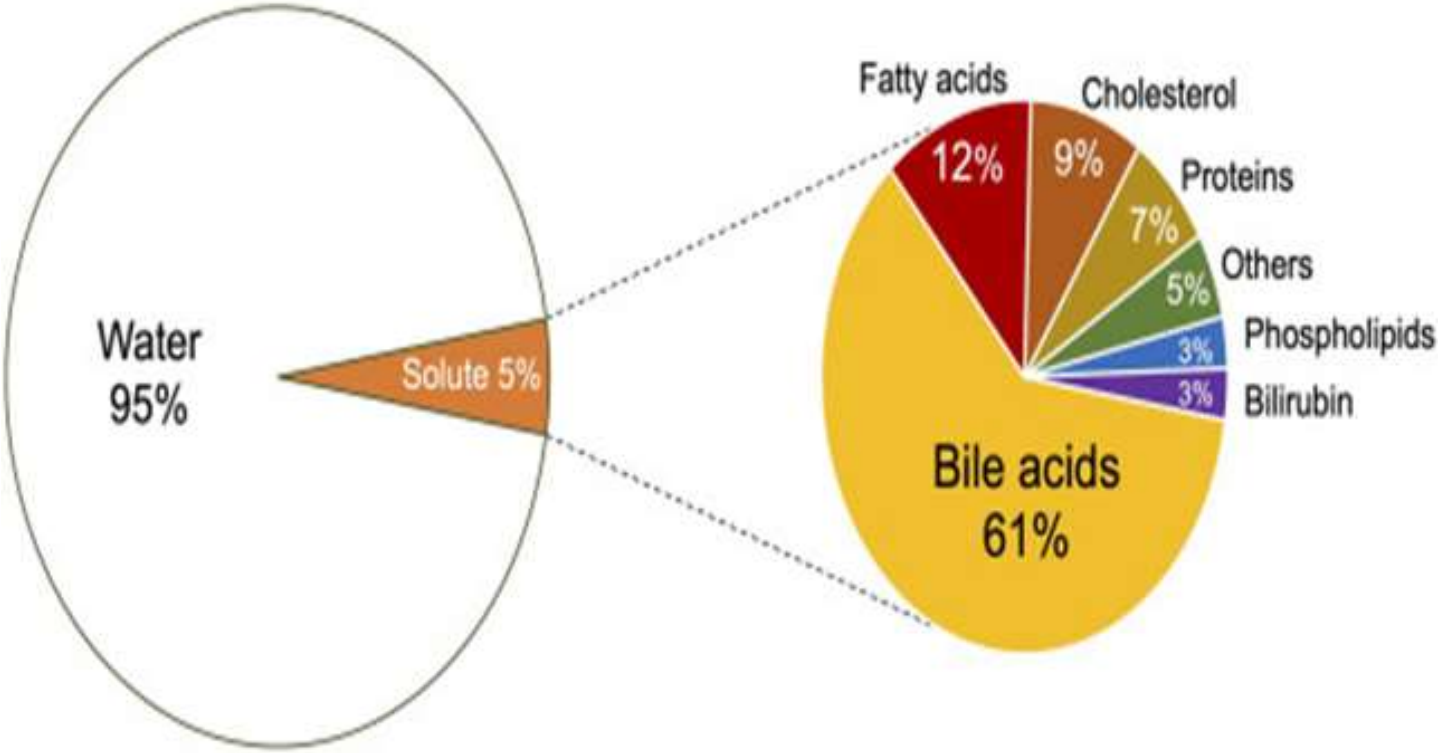
Bile (synthesis, composition and function)

- **Bile is a greenish-yellow fluid produced by hepatocytes in the liver and stored in the gallbladder, essential for digesting fats, absorbing fat-soluble vitamins (A, D, E, K), and excreting waste like bilirubin and cholesterol. It is composed primarily of water, bile salts, phospholipids, cholesterol, and bile pigments**

- **1. Bile is a complex, aqueous mixture consisting of approximately 95% water and 5% solutes.**
- **2. Bile Salts/Acids: The main functional component, essential for digestion.**
- **3. Phospholipids: Primarily lecithin, which helps solubilize cholesterol.**

- **4. Cholesterol: Excreted as a waste product.**
- **5. Bile Pigments: Primarily conjugated bilirubin, which gives bile its color.**
- **6. Electrolytes: Sodium, potassium, calcium, and bicarbonate (creates an alkaline pH)**

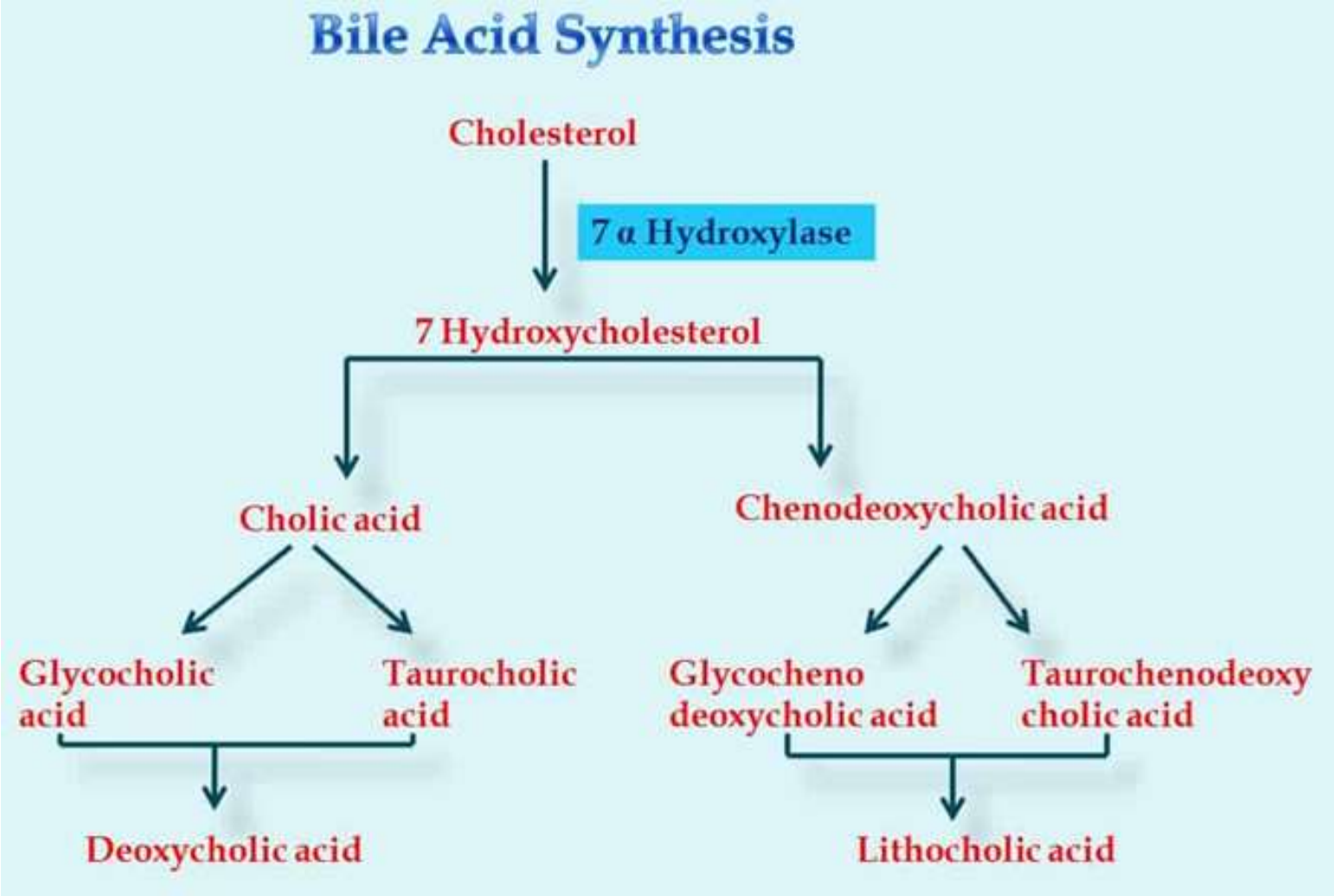
Bile Composition

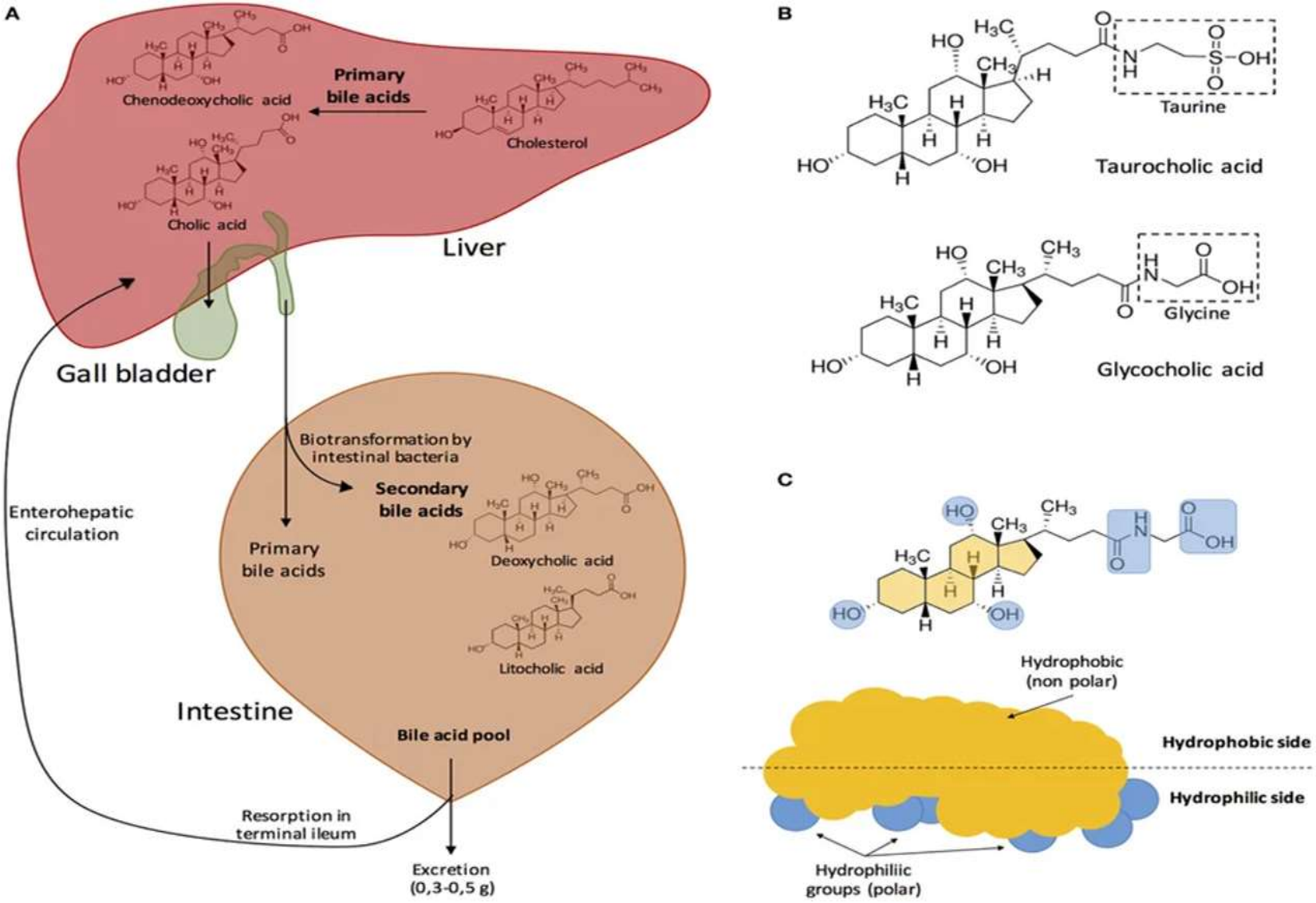


Bile acid Synthesis and Metabolism

- **Bile synthesis occurs in the liver, where hepatocytes convert cholesterol into primary bile acids (cholic acid and chenodeoxycholic acid).**
- **1. Conjugation: These bile acids are conjugated with amino acids (taurine or glycine) to form bile salts, which increases their solubility and effectiveness.**

- **2. Enterohepatic Circulation: About 95% of bile salts are reabsorbed in the terminal ileum and returned to the liver through the portal vein, allowing them to be reused multiple times.**
- **3. Secondary Bile Acids: Bacteria in the colon convert primary bile acids into secondary bile acids, such as deoxycholic acid and lithocholic acid**


Bile Acid Synthesis
Cholesterol
7 α Hydroxylase
7 Hydroxycholesterol
Cholic acid
Chenodeoxycholic acid
Glycocholic acid
Taurocholic acid
Glycocheno deoxycholic acid
Taurochenodeoxy cholic acid
Deoxycholic acid
Lithocholic acid


A
Chenodeoxycholic acid
Primary bile acids
Cholesterol
Cholic acid
Liver
Gall bladder
Biotransformation by intestinal bacteria
Secondary bile acids
Enterohepatic circulation
Primary bile acids
Deoxycholic acid
Litocholic acid
Intestine
Bile acid pool
Resorption in terminal ileum
Excretion (0,3-0,5 g)
B
Taurine
Taurocholic acid
Glycine
Glycocholic acid
C
Hydrophobic (non polar)
Hydrophobic side
Hydrophilic side
Hydrophiliic groups (polar)

Functions of Bile

- **Bile serves several key digestive and excretory functions:**
- **1. Fat Emulsification: Breaks down large fat globules into small droplets (micelles), significantly increasing the surface area for lipase enzymes to digest fat.**
- **2. Absorption of Fats: Facilitates the absorption of lipids and fat-soluble vitamins (A, D, E, and K) from the intestine.**

- **3. Excretion: Provides a route for the elimination of excess cholesterol, bilirubin (from broken-down red blood cells), toxins, and heavy metals (copper, zinc).**
- **4. Neutralization: Its alkaline pH neutralizes acidic chyme entering the duodenum from the stomach.**
- **5. Immune Defense: Contains IgA and aids in killing harmful bacteria in the intestine**

Gallstones formation

- **Gallstones (cholelithiasis) form primarily due to cholesterol supersaturation in bile, gallbladder stasis, and crystal nucleation. Excess hepatic cholesterol secretion leads to sludge formation, which progresses to solid stone growth.**

- **Types of Gallstones & Pathogenesis**
- **1. Cholesterol Stones (approx. 80%): Develop in sterile bile when cholesterol exceeds the capacity of bile acids and phospholipids to keep it soluble, forming cholesterol crystals.**
- **2. Black Pigment Stones: Composed of calcium bilirubinate. Form in sterile bile, often linked to hemolytic anemia (increased bilirubin) or liver cirrhosis.**

- **3-Brown Pigment Stones: Formed due to bacterial infection, which hydrolyzes biliary lipids, causing calcium soaps and bilirubinate to precipitate. These are associated with biliary infection and stasis**

Absorption of nutrients

- **Absorption can occur through five mechanisms: (1) active transport, (2) passive diffusion, (3) facilitated diffusion, (4) co-transport (or secondary active transport), and (5) endocytosis**

Malabsorption syndromes

- **I-Fat Malabsorption**
- **Etiologies of Fat Malabsorption**
- **1. Decreased duodenal pH: An optimal duodenal pH of 6.5 is necessary for effective fat digestion. Zollinger-Ellison syndrome lowers duodenal pH due to excessive gastric acid secretion, which inactivates pancreatic enzymes.**

2-Loss of absorptive intestinal surface area:

3-Impaired lipid processing: Impaired lipid processing by bile acids results when bile acid synthesis remains insufficient for effective fat absorption.

4-Small intestinal bacterial overgrowth

5-**Exocrine pancreatic insufficiency:** results from defective production of pancreatic lipase, colipase, and bicarbonate. Chronic pancreatitis, frequently associated with alcohol use disorder or prolonged biliary obstruction

II-Carbohydrate Malabsorption

- **Etiologies of Carbohydrate Malabsorption**
- **impaired enzymatic breakdown: A deficiency in pancreatic amylase can hinder the hydrolysis of complex carbohydrates into disaccharides. Inadequate disaccharidase activity further compromises carbohydrate digestion**

- **reduced absorptive capacity**
- **structural abnormalities of the small intestine.**
- **Inadequate disaccharidase activity: lactase deficiency representing the most common type,(Lactase, an enzyme found on the surface of small intestinal microvilli breaks down lactose into glucose and galactose)**

Lactose intolerance

- **the inability to fully digest lactose (milk sugar) due to low lactase enzyme levels, leading to bloating, diarrhea, abdominal pain, and gas. Adult-onset lactase deficiency affects the majority of the global population.**

Types Lactose intolerance

1. Primary (Most Common): Develops after childhood as lactase production drops.
2. Secondary: Occurs after illness, injury, or damage to the small intestine.
3. Congenital/Developmental: Present from birth or due to premature birth

- **Treatment**
- **Dietary Changes: Limiting milk and dairy products.**
- **Lactase Supplements: Taking drops or tablets to aid digestion.**
- **Lactose-Free Alternatives: Using lactose-free milk, soy, or almond products.**
- **Dairy Substitutes: Some yogurt or hard cheeses might be better tolerated than milk**

III-Protein Malabsorption

- **Etiologies of Protein Malabsorption**
- **Protein malabsorption may result from impaired secretion or activity of pancreatic substances, particularly bicarbonate and proteases. Conditions, eg, chronic pancreatitis and cystic fibrosis, often disrupt normal enzyme production or delivery, thereby limiting the breakdown of dietary proteins into absorbable amino acids and peptides.**

Quiz

- **1. Regarding carbohydrate absorption and digestion**
- A. Salivary amylase has an optimal pH of 8.7
- B. GLUT4 is insulin dependent.
- C. GLUT3 is responsible for most glucose uptake
- D. Glucose and fructose are transported by the same carrier

- **2. which of The following proteolytic enzyme is produced by the stomach?**

A. Trypsin

B. Chymotrypsin

C. Elastase

D. pepsin

- **3- What is responsible for bile production?**

A. Liver

B. Gallbladder

C. Small intestines

D. Large intestines

. What is the main waste product excreted in bile?

A. Bilirubin.
B. Glucose
C. bile salts
D. Albumin

What is the main function of bile in the digestive system?

A. Regulation of body temperature

B. Transport of oxygen in the blood

C. Emulsification and digestion of fats

D. Production of insulin

- **A 55-year-old woman presents with acute onset abdominal pain radiating to her back, nausea, and vomiting. CT scan suggests a diagnosis of acute pancreatitis. The pathogenesis of acute pancreatitis relates to inappropriate activation of trypsinogen to trypsin. Which of the following activates trypsin in normal digestion?**

A. Secretin

B. Lipase

C. Cholecystokinin

D. Enterokinase

What is the composition of bile?

A. Water, bile salts, cholesterol, and bilirubin

B. Water, sodium, potassium, and chloride

C. Protein, carbohydrates, fats, and vitamins

D. Calcium, magnesium, phosphorus, and zinc